P2: Plotting Oceanographic Data And Timeseries

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SEA4001W

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1. Temperature-Salinity Plot:

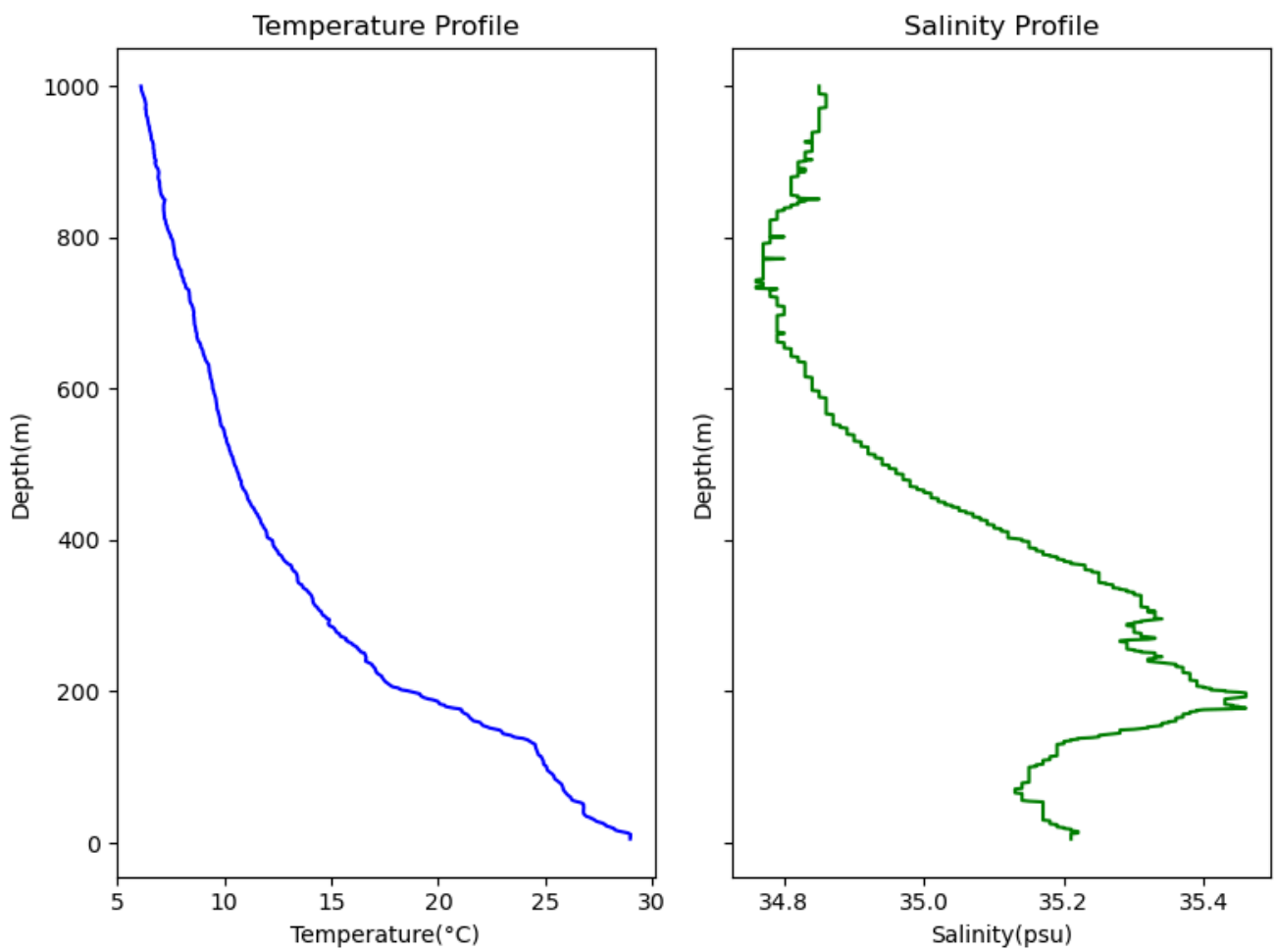

Figure 1 : Temperature (°C) and Salinity (psu) profiles with shared Y-axis (Depth(m)).

Prompt for ChatGPT to help me improve my plot: Chat, this is my code:

1 hidden cell

Continued Prompt: How can I improve my plot for future references?

ChatGPT Response:

1. Invert the y-axis (Depth (m))
2. Remove Duplicate Y-axis
3. Use grid line

```
ax[0].invert_yaxis()
```

```
ax[0].set_ylabel('Depth (m)')
```

```
ax[0].grid(True)
ax[1].grid(True)
```

2. Temperature plot

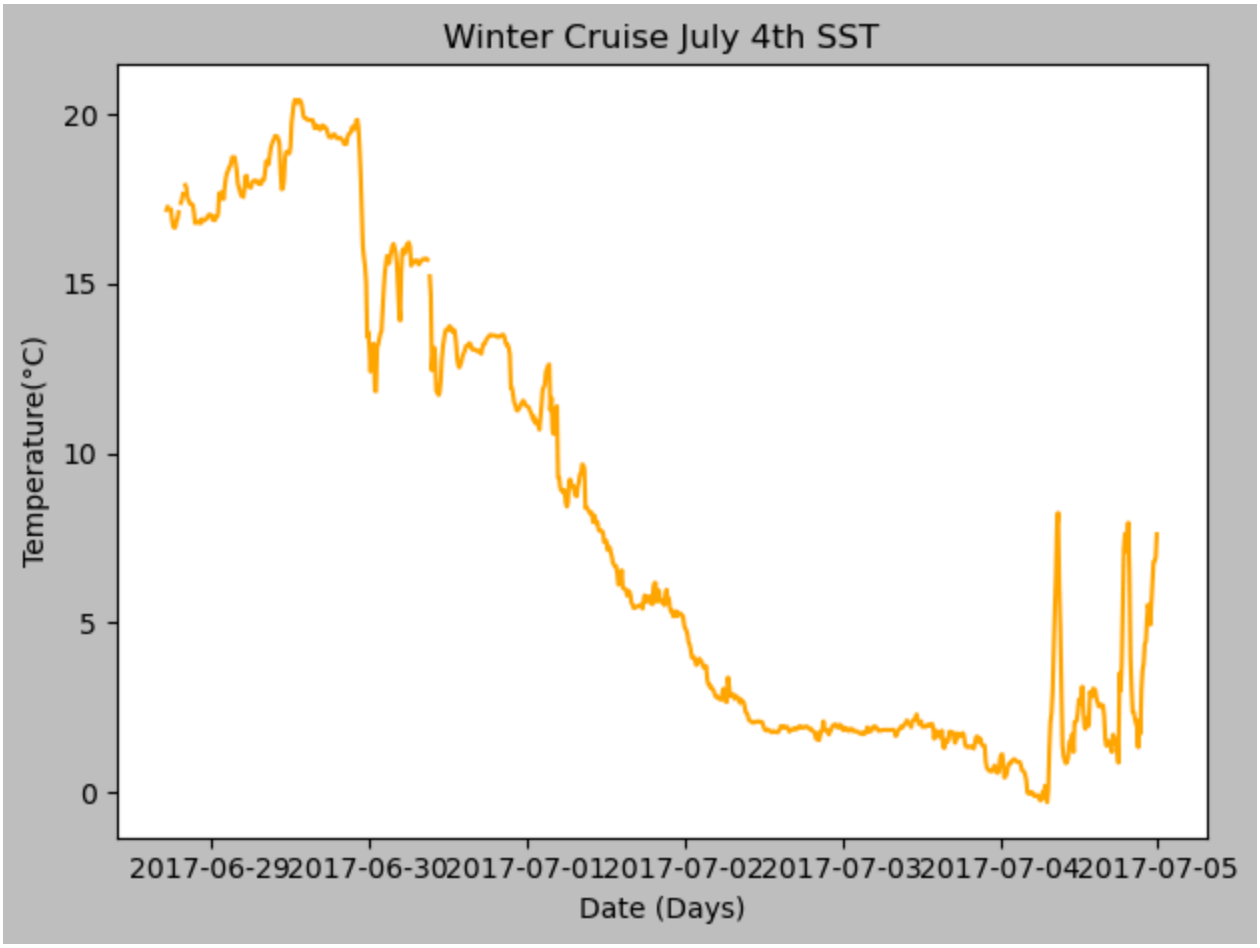

Figure 2 : Temperature (°C) of the Winter Cruise in Time (Days).

Prompt for ChatGPT to help me improve my plot: Chat, this is my code:

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Continued Prompt: How can I improve my plot for future references?

1. Improve the title: Sea Surface Temperature – Winter Cruise (4 July 2017)
2. Add gridline
3. Format x-axis better

```
ax.grid(True)
```

```
fig.autofmt_xdate()
```

3. Histogram Plot

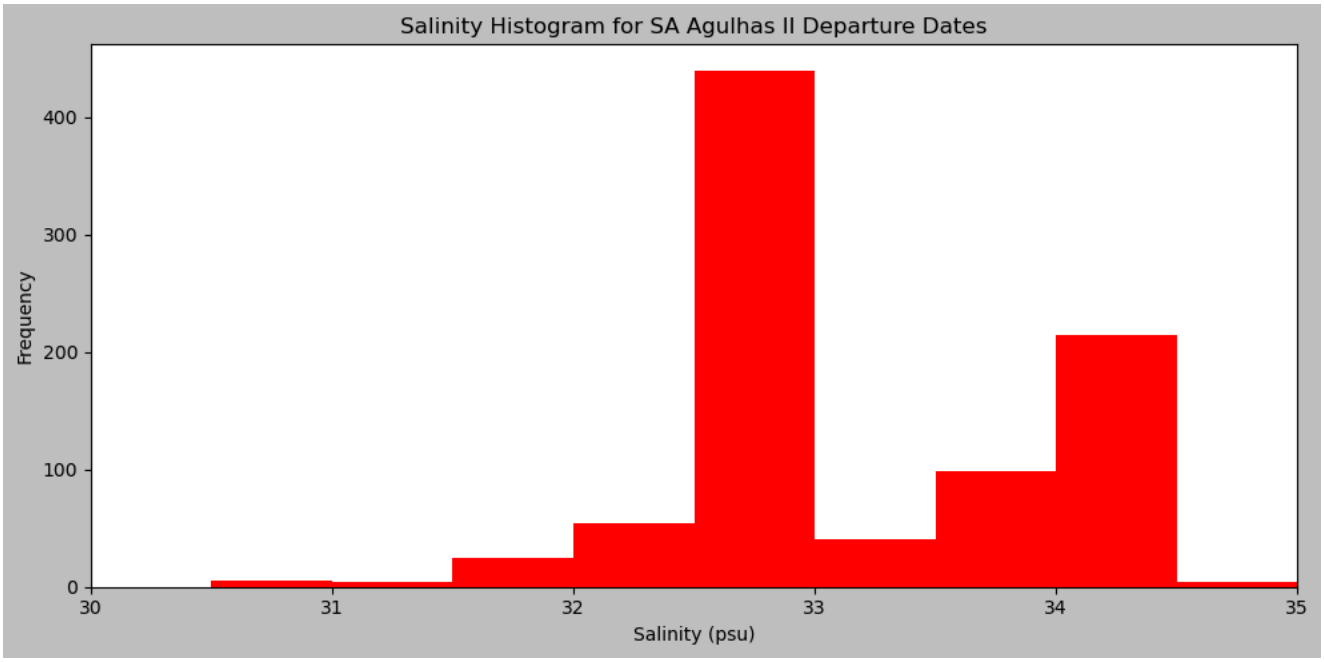

Figure 3 : Histogram of the SA Agulhas II Winter Cruise.

Prompt for ChatGPT to help me improve my plot: Chat, this is my code:

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Continued Prompt: How can I improve my plot for future references?

1. Add mean line for better Data Analysis

```
mean_sal = july_df['TSG_SALINITY'].mean()
ax.axvline(mean_sal, color='black', linestyle='--', label=f'Mean = {mean_sal:.2f}')
ax.legend()
```

4. Scatter Plot

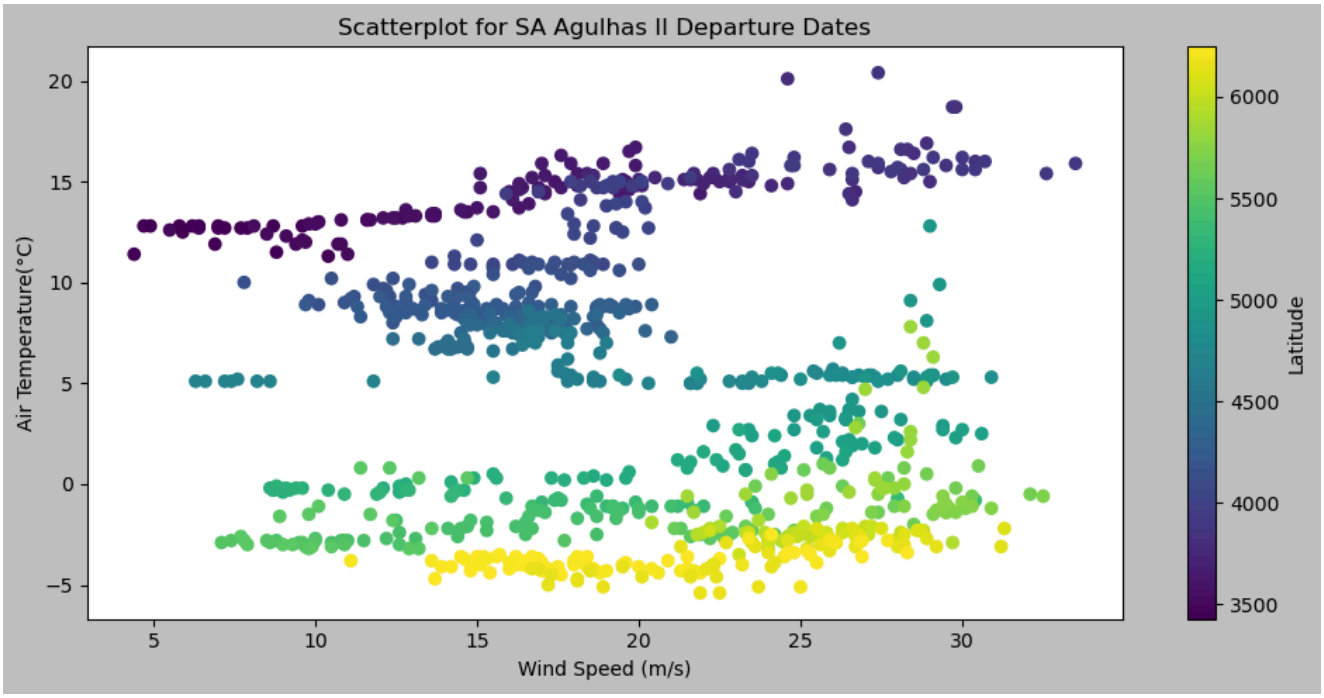

Figure 4 : Scatterplot of the SA Agulhas II Winter Cruise Air Temperature (°C) vs Wind Speed (m/s).

Prompt for ChatGPT to help me improve my plot: Chat, this is my code:

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1. Add gridlines
2. Add transparency argument

```
ax.grid(True)
```

```
scatter = ax.scatter(..., alpha=0.7)
```

5. Statistical Table:

```
                 Mean  ... Interquartile Range
Temperature   8.146541  ...           11.768850
Salinity     33.110553  ...            1.272575

[2 rows x 3 columns]
```

Figure 5 : Statistical table of the SA Agulhas II Winter Cruise of Temperature (°C), Salinity (psu) and IQR.